

# Nikita Kazeev

[kazeevn@gmail.com](mailto:kazeevn@gmail.com) · [Website](#) · [GitHub](#) · [LinkedIn](#) · [Google Scholar](#) · [ORCID](#)

*Research Scientist at the intersection of AI and Physics*

## WORK EXPERIENCE

**CONSTRUCTOR GROUP** *Machine Learning Consultant* 2021–present

**NATIONAL UNIVERSITY OF SINGAPORE** *Postdoc under [Kostya Novoselov](#)* 2022–present

**Wyckoff Transformer, a generative model for materials** [ICML 2025](#) · [GitHub](#)

- Developed a coordinate-free permutation-invariant autoregressive Transformer encoder model for material generation and property prediction based on symmetry inductive bias
- Increased materials discovery yield (S.U.N.) by 1.24x
- Accelerated inference by 4 orders of magnitude, to 0.05 GPU ms per structure
- Implemented the model in PyTorch; production-ready package published on [PyPI](#)
- Orchestrated large-scale (10k structures) DFT computations with VASP at an HPC cluster
- Led a team of 6

**ML for defects in 2D crystals** [npj 2D Mater. Appl.](#) · [npj Comput. Mater.](#) · [patent](#) · [GitHub](#)

- Conceived a novel sparse representation of crystals with defects
- Achieved a 3.7x energy prediction error reduction
- Optimised training and inference: 4x less memory, 8x fewer GPU operations
- Developed code for reproducible parallel running of machine learning experiments
- Led a team of 3

**(CERN | YANDEX → HSE UNIVERSITY)** *Researcher* 2014–2022

[2025 Breakthrough Prize in Fundamental Physics](#) as a member of LHCb.

**Generative models for fast simulation** [paper 1](#) · [paper 2](#) · [paper 3](#) · [code](#)

- Developed a GAN-based machine learning model for high-fidelity simulation of a Cherenkov detector trained on tabular data.
- Achieved approximately  $10^5$  speed-up compared to the ab-initio simulator.

**Machine learning on noisy data** [paper 1](#) · [paper 2](#) · [code](#)

- Developed a rigorous way to train ML algorithms on data with the label-noise model common in high-energy physics.

**Muon identification at the LHCb experiment at CERN** [paper](#) · [poster](#) · [IDAO-2019](#)

- Solved a classification problem over tabular data with noisy labels under timing constraints.
- Developed a gradient boosting model that reduced the false positive rate by 30% in the critical low-track-momentum region.
- Integrated the model into LHCb production, mostly in C++.
- Packaged the work as a data science competition problem for IDAO-2019.

## EDUCATION

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| <b>HSE University &amp; Sapienza Università di Roma</b><br><i>PhD in Computer Science and Physics (Double Degree)</i><br>Supervisors: <a href="#">Andrey Ustyuzhanin</a> & <a href="#">Barbara Sciascia</a><br>Thesis: <a href="#">Machine Learning for particle identification in the LHCb detector</a> | 2016–2020   |
| <b>Product Management course at Yandex</b><br><i>Focused coursework in product strategy and execution.</i>                                                                                                                                                                                               | 2018        |
| <b>Yandex School of Data Analysis</b><br><i>Master's-level CS programme</i><br>Algorithms, machine learning, deep learning, and distributed systems.                                                                                                                                                     | 2013–2015   |
| <b>Moscow Institute of Physics and Technology</b><br><i>MS in Physics</i><br>Optimisation of data processing of the LHCb experiment.                                                                                                                                                                     | 2014–2016   |
| <i>BS in Physics</i><br>Wave-packet molecular dynamics of electrons in nonideal plasma.                                                                                                                                                                                                                  | 2010–2014   |
| <b>International Junior Science Olympiad (IJSO)</b><br>Silver medal                                                                                                                                                                                                                                      | Korea, 2008 |

## MENTORSHIP

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- Mentored 9 students ([1](#), [2](#), [3](#), [4](#), [5](#), [6](#), [7](#), [8](#), [9](#)), 3 interns, and student workshop projects ([1](#), [2](#)).
- Co-PI of a \$3.4m AI Singapore grant on multiscale machine learning.

## TEACHING & OUTREACH

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- Pitch research to the general public through major newspaper coverage ([1](#), [2](#)), blog posts ([1](#), [2](#)), a [science slam](#), interviews ([1](#), [2](#)), and a public [talk](#).
- Taught at the Machine Learning in High Energy Physics Summer School in [2015](#), [2016](#), [2017](#), [2018](#), [2019](#), [2020](#), [2021](#), and [2023](#).
- Lecturer in the [award-winning](#) Coursera Advanced Machine Learning [specialisation](#)
- Delivered more than 20 [conference talks](#).

## SERVICE

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- Reviewer for NeurIPS, RSC Advances, Machine Learning: Science and Technology, Digital Discovery
- Peer Staff Supporter at NUS, serving as a first-line support for mental wellbeing.

## SKILLS

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- **Generative modelling for scientific data** – GANs, autoregressive transformers, and diffusion; from fast detector simulation to symmetry-aware crystal generation.
- **Uncertainty quantification** – built the first methods for estimating the uncertainty of conditional GANs; calibrated generative and discriminative models.
- **ML on structured, non-text data** – tabular, graph, and atomistic representations, spanning gradient boosting to transformers.
- **Inference and systems at scale** – C++ production integration, large-scale DFT/VASP on HPC clusters, and reproducible experiment pipelines.
- **Physics domain depth** – high-energy and condensed-matter physics; first-principles intuition that shapes model design.
- **Research leadership** – lead multidisciplinary teams, chair conferences, and mentor students from idea to publication.

ML & AI • Python • PyTorch • C++ • HPC • Physics • Research Writing & Speaking • Leadership & Mentorship

## PRESENTING AT ICML 2026

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July 11, 12:15–13:30 – Nikita Kazeev and Ian Babich. [“Foresight-Phys: A Benchmark for Forecasting the Results of Physical Experiments”](#). In: *Forecasting as a New Frontier of Intelligence Workshop at ICML 2026*.

July 11, 10:50–11:50 & 15:55–16:55 – Artem Dembitskiy\*, Shuya Yamazaki\*, Artem Maevskiy\*, **Nikita Kazeev\***, Roman A. Eremin, Semen Budennyi, Kedar Hippalgaonkar, Antonio Helio Castro Neto, and Andrey E. Ustyuzhanin. [“Generative Pipeline for Discovering Solid-State Battery Materials with Universal Atomistic Potentials”](#). In: *AI for Science Workshop at ICML 2026*. \*Equal Contribution.

## SELECTED PUBLICATIONS

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Nikita Kazeev, Wei Nong, Ignat Romanov, Ruiming Zhu, Andrey E Ustyuzhanin, Shuya Yamazaki, and Kedar Hippalgaonkar. [“Wyckoff Transformer: Generation of Symmetric Crystals”](#). In: *Proceedings of the 42nd International Conference on Machine Learning*. Ed. by Aarti Singh, Maryam Fazel, Daniel Hsu, Simon Lacoste-Julien, Felix Berkenkamp, Tegan Maharaj, Kiri Wagstaff, and Jerry Zhu. Vol. 267. Proceedings of Machine Learning Research. PMLR, 13–19 Jul 2025, pp. 29495–29526.

Nikita Kazeev, A. R. Al-Maeeni, I. Romanov, et al. [“Sparse representation for machine learning the properties of defects in 2D materials”](#). In: *npj Computational Materials* 9 (2023), p. 113.

Denis Derkach, Nikita Kazeev, Fedor Ratnikov, Andrey Ustyuzhanin, and Alexandra Volokhova. [“Cherenkov detectors fast simulation using neural networks”](#). In: *Nuclear Instruments and Methods in Physics Research Section A* (2019). Alphabetic order, I’m the corresponding author.